

Howard University
College of Engineering & Architecture
Department of Electrical Engineering & Computer Science
EECE 306-01: Fundamentals of Electromagnetics Laboratory
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Lab 5: Flux, Gauss's Law, and the Divergence

Objectives

By the end of this session you will have

- implemented a closed surface integrator that works on any field,
- turned Gauss's law from a theorem into a test your code must pass,
- confirmed that the answer does not depend on the shape of the surface,
- verified the point form of Gauss's law with a numerical divergence.

1. Three functions, and a change of habit

Only three functions are due today, because the integrator built last week does most of the work. What changes today is not the amount of code but the way it is used.

Until now, every test compared a computed number against something. Today the test is a *law*. Gauss's law says that the closed surface integral of $\vec{D}$ equals the enclosed free charge, for any charge distribution and any closed surface. Your code either satisfies that or it does not, and no reference answer is required to find out.

2. Part I. `em.field.D`

$$\vec{D} = \epsilon_r \epsilon_0 \vec{E}$$

A thin wrapper on `em.field.E`, with `epsr` defaulting to 1. It exists as a separate function because Gauss's law is a statement about $\vec{D}$ and not about $\vec{E}$, and because in Lab 8 the permittivity stops being a constant.

3. Part II. `em.field.flux`, and why it takes a function handle

$$\Phi = \oint_S \vec{F} \cdot d\vec{S}$$

The signature is `Phi = em.field.flux(F, surf, uspan, vspan, Nu, Nv, rule)` where **F** is a **field function handle**, not a source struct. The call site looks like this.

```
s = em.src.pointCharge(2e-9, [0 0 0]);
F = @(r) em.field.D(s, r);
sph = @(th,ph) 0.5*[sin(th).*cos(ph), sin(th).*sin(ph), cos(th)];
Phi = em.field.flux(F, sph, [0 pi], [0 2*pi], 80, 160);
```

The reason for the handle rather than the source is that from Lab 7 onward you will pass fields that have *no source struct behind them*. The output of the Poisson solver is an array of potential values on a grid. Wrapped in an interpolant it becomes a function handle, and it flows into this

same integrator with no special case. A version of `em.field.flux` that only accepts sources would have to be rewritten in five weeks.

The vector area element comes from the same cross product that produced the surface weights last week, but now you need its *direction* as well as its magnitude,

$$d\vec{S} = \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du dv$$

The orientation follows from the order of the two parameters. Getting it backwards flips the sign of every flux you ever compute, so check the sign once, on a case where you know the answer, and record your convention in the help block.

4. Part III. Gauss's law as a test

4.1 The enclosed case

Put a known charge inside a spherical surface and integrate.

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{\text{enc}}$$

Assert equality with a relative tolerance. Then move the same charge *outside* the surface without changing anything else, and assert the integral is zero, this time with an absolute tolerance for the reasons discussed in Lab 3.

The most powerful test in the electrostatics half of this course

Look at what this test does not need. It does not need an analytic expression for the field. It does not need a symmetric charge distribution. It does not need a symmetric surface. It works for a single point charge, for a hundred point charges at random positions, for a curved line charge, and for a volume charge with a nonuniform density, all with the same three lines of code.

That is what makes a conservation law valuable as a verification tool. Every other test you have written so far has been tied to a specific configuration. This one is tied to nothing, so it can be run on the real problem rather than on a simplified stand-in.

4.2 The shape independence check

Repeat the enclosed case with three different closed surfaces around the same charge.

1. A sphere.
2. A cube, parametrized as six square faces summed together.
3. A hemisphere closed by its equatorial disk, again two pieces summed.

All three must give the same answer, because Gauss's law does not care about the shape of the surface. Report the three numbers side by side.

This is a stronger statement than it first appears. If the cube and the sphere disagree, the physics is not at fault, because the physics is the same in both cases. The fault is in your quadrature, most likely in an outward normal that points inward on one face, or in a piecewise parametrization whose pieces do not join. Note in your notebook which of the three was hardest to get right and why.

4.3 Superposition and the enclosed charge

Build a source with several point charges, some inside the surface and some outside, using `em.src.merge`. The flux must equal the sum of only the enclosed ones. This tests both the integrator and your understanding, and it is the case students most often predict incorrectly on an exam.

5. Part IV. The point form and the divergence

Implement `em.op.div(F, r, h)` by central differences,

$$\nabla \cdot \vec{F} \approx \sum_{k=1}^3 \frac{F_k(\vec{r} + h\hat{e}_k) - F_k(\vec{r} - h\hat{e}_k)}{2h}$$

The point form of Gauss's law states

$$\nabla \cdot \vec{D} = \rho_v$$

Two checks.

Inside a uniform volume charge. Build a spherical volume charge of uniform density ρ_v and evaluate the divergence of $\vec{D}$ at points well inside it. The result must equal ρ_v . Be careful about the discretization, since a coarse element grid makes the interior field lumpy at the scale of the element spacing, and the divergence will pick that up. Refine until the answer stabilizes, and report how many elements were needed.

In a charge-free region. Evaluate the divergence of $\vec{D}$ outside any source. It must vanish. This is the differential statement of the same law you just verified in integral form, and confirming both is a consistency check on your own code.

The step size

The parameter `h` defaults to 10^{-5} m. Take the default for today. Next week explains where it comes from, and the short version is that driving h smaller does not keep making the derivative better.

6. Part V. A note on speed

From this lab onward, vectorization is no longer optional. A closed surface integral with $N_u = 80$ and $N_v = 160$ evaluates the field at 12800 points, and a single lab session runs several of these. If your `em.field.E` still loops over observation points, fix it today rather than in Lab 8 when the solver makes it ten times worse.

Required tests

Write `tests/test_lab05.m` containing at least these checks.

1. Flux of $\vec{D}$ through a sphere enclosing a point charge equals the charge, to a relative tolerance of 10^{-6} .
2. The same charge moved outside the sphere gives a flux consistent with zero, absolute tolerance justified in your notebook.
3. Sphere, cube and hemisphere-plus-disk agree with each other to a relative tolerance of 10^{-4} .
4. A merged source with three charges inside and two outside gives the sum of the three inside.

5. Flux through the sphere is independent of the charge position while it remains inside, checked at three off-center positions.
6. Divergence of $\vec{D}$ inside a uniform volume charge equals ρ_v within five percent at your reported element count.
7. Divergence of $\vec{D}$ in a charge-free region is consistent with zero.
8. Reversing the parameter order in the surface handle flips the sign of the flux, confirming that you understand your own orientation convention.

Deliverables

Submit through Canvas by the start of the next lab session.

1. **Toolbox snapshot**, `TeamMN_Lab05.zip`.
2. **Lab notebook**, published script exported to PDF, three to five pages, with the three surface results side by side, your account of which surface was hardest and why, and your element count study for the divergence.
3. **Test output** pasted into the notebook.

Looking ahead

Next week brings the electric potential, the numerical gradient, and a second conservation law to test against. It also brings the study of finite difference step size promised above. Read Section 2.9 of Wentworth before you come.